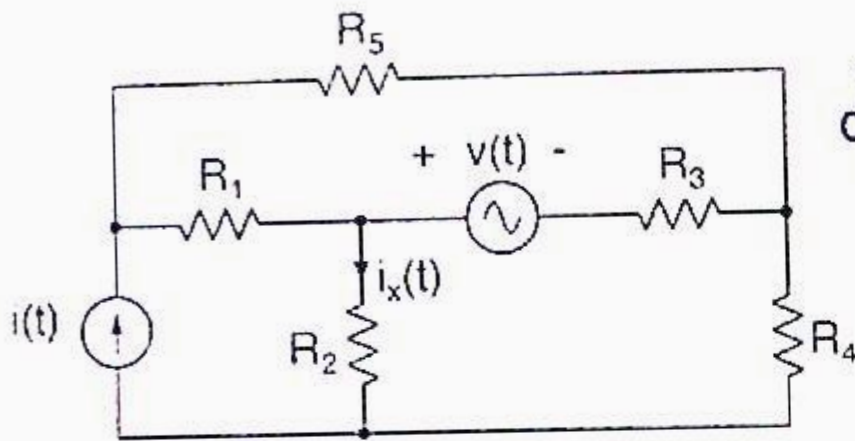


MIDTERM EXAM - SOLUTIONS

Question 1 [25 Points]:

For the circuit below

- 1 a) Mark all the nodes in the circuit
- 1 b) Select a common node, and mark it as ground node.
- 5 c) Label the node voltages as V_A, V_B .
- 5 d) Write down the node equations at every node (except the common node)
- 2 e) Find the node voltages if $R_1 = R_2 = R_3 = R_4 = R_5 = 1 \text{ k}\Omega$, and $i(t) = \cos(\omega t) \text{ mA}$ and $v(t) = \cos(\omega t) \text{ V}$
- 2 f) For the same values in part (e), find the current $i_x(t)$.



d) Node A: $\frac{V_A - V_B}{R_1} + \frac{V_A - V_C}{R_5} - i = 0$

Node B: $\frac{V_B - V_A}{R_1} + \frac{V_B - V - V_C}{R_3} + \frac{V_B}{R_2} = 0$

Node C: $\frac{V_C - V_A}{R_5} + \frac{V_C}{R_4} + \frac{V_C + V - V_B}{R_3} = 0$

$R_1 = R_2 = R_3 = R_4 = R_5 = R$

e) $2V_A - V_B - V_C = iR$

$3V_B - V_A - V_C = V$

$3V_C - V_A - V_B = -V$

$\Rightarrow$

$V_C = iR - V_B \Rightarrow V_A = iR + V_B$

$V_B = \frac{1}{4}V + \frac{1}{2}iR$

$V_C = \frac{1}{2}iR - \frac{1}{4}V$

$\therefore 2V_A$

$\therefore V_A(t) = (1 \text{ mA})(1 \text{ k}\Omega) \cos \omega t$
 $= 1 \text{ V} \cos \omega t$

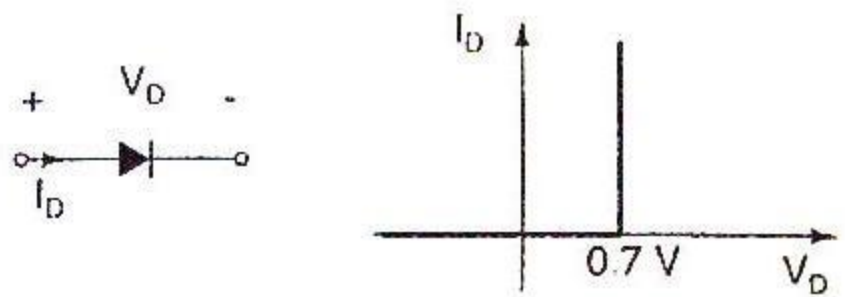
$V_B(t) = \frac{3}{4} \text{ V} \cos \omega t$

$V_C(t) = \frac{1}{4} \text{ V} \cos \omega t$

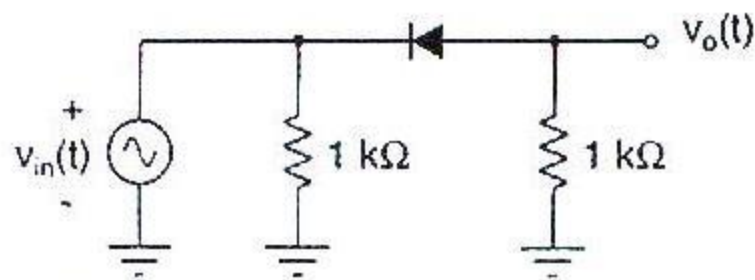
f) $i_x = \frac{V_B}{1 \text{ k}} = \frac{3}{4} (\text{mA}) \cos \omega t$

Question 2 [25 points]:

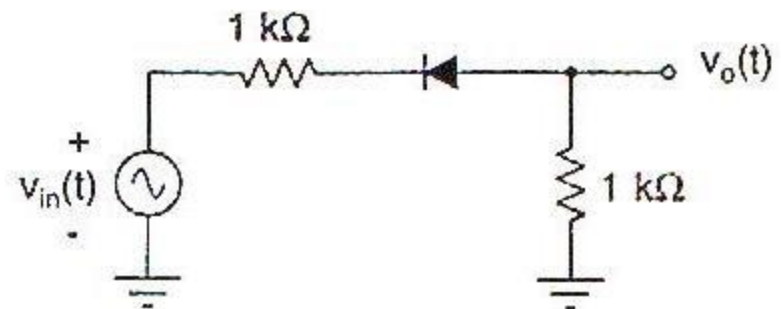
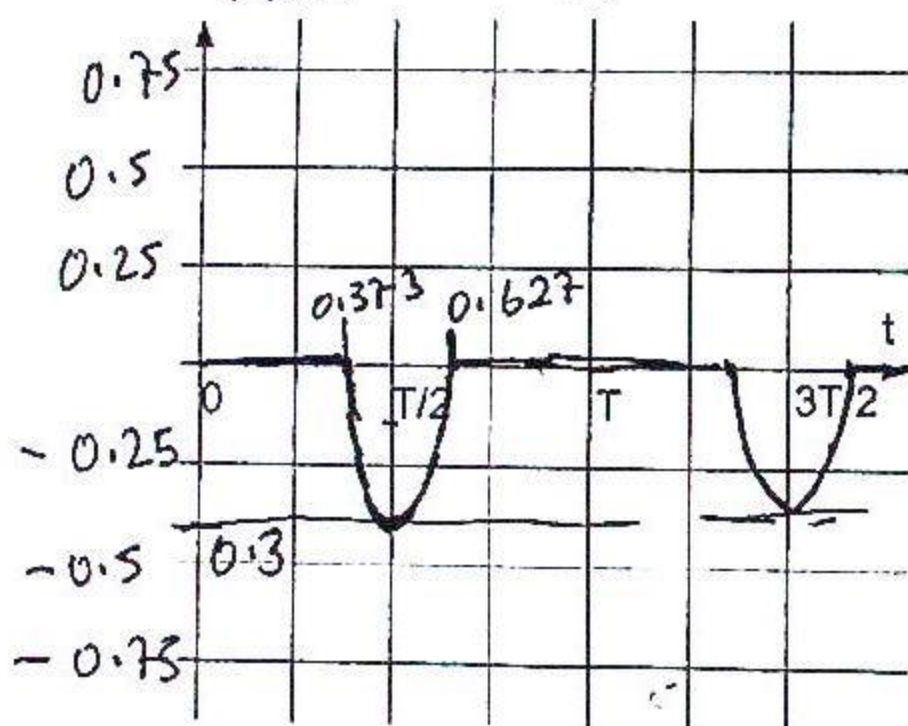
For the circuits in (a) and (b) below, the diode has the voltage-current relationship shown on the right.



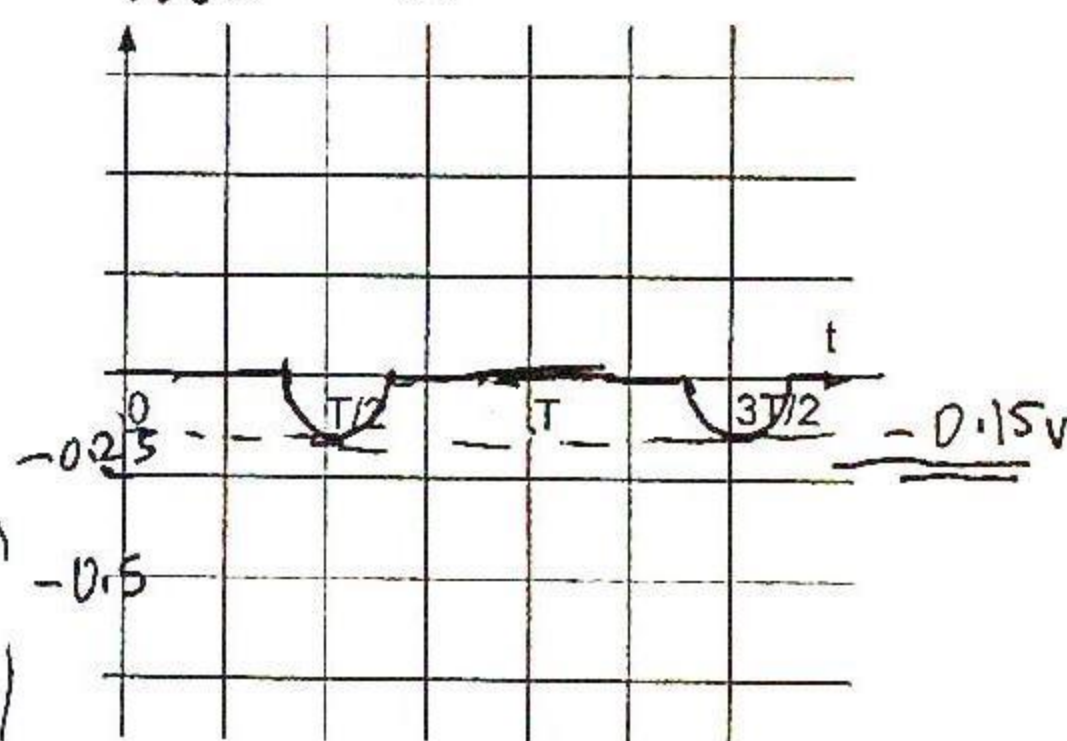
For both the circuits in (a) and (b), find $v_o(t)$ if $v_{in}(t) = \cos(\omega t) V$. Show all your calculations, justify your assumptions (if any) clearly. Label and scale the vertical axis and plot the waveforms on the provided grids, clearly indicating all critical points.



(a)



(b)



$$\begin{aligned}
 v_{in} > -0.7 &\Rightarrow \text{Diode OFF} \Rightarrow v_o(t) = 0 \\
 v_{in} < -0.7 &\Rightarrow \text{Diode ON} \Rightarrow v_o(t) = v_{in}(t) + 0.7 \\
 v_{in}(t) = -0.7 &\Rightarrow \cos \omega t = -0.7
 \end{aligned}$$

$$\begin{aligned}
 \omega t &= \frac{2\pi}{T} t = \cos^{-1}(-0.7) \\
 &= \{ 134.4^\circ = 2.35 \text{ rad} \}
 \end{aligned}$$

$$\frac{t}{T} = \begin{cases} 0.373 \\ 0.627 \end{cases}$$

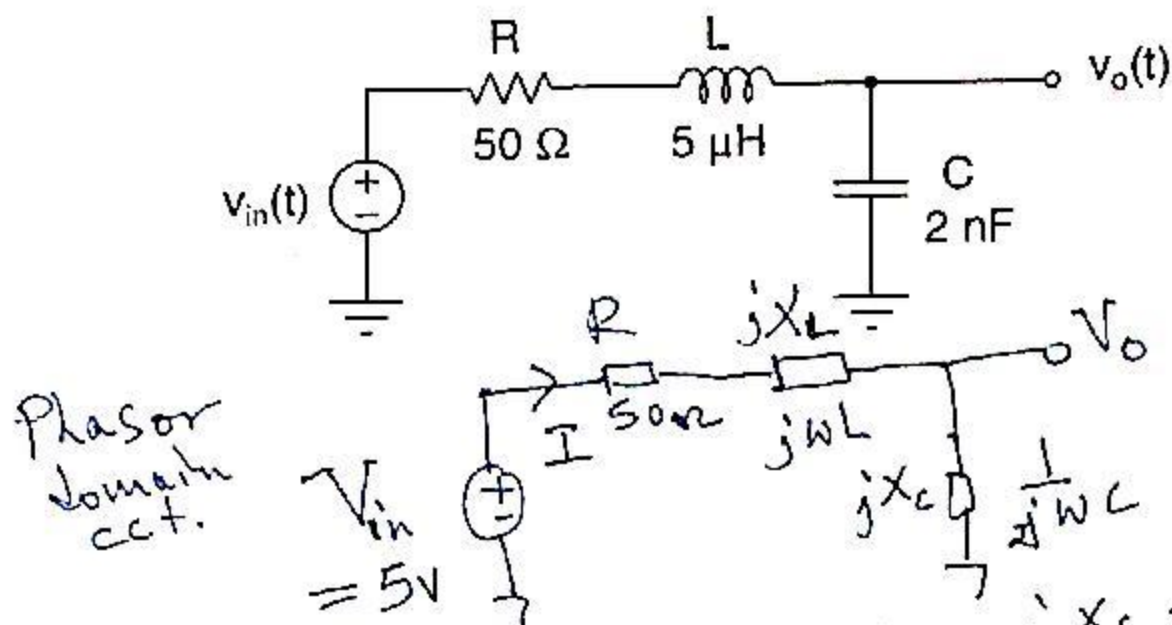
$$\begin{aligned}
 v_{in} > -0.7 &\Rightarrow v_o(t) = 0 \\
 v_{in} < -0.7 &\Rightarrow v_o(t) = (v_{in} + 0.7) \frac{1k}{1k+1k} \\
 &= \frac{1}{2} (v_{in}(t) + 0.7)
 \end{aligned}$$

$$\Rightarrow \checkmark \quad \frac{t}{T} = \begin{cases} 0.373 \\ 0.627 \end{cases}$$

Question 3 [25 points]:

For the circuit below:

- a) Find $v_o(t)$ if $v_{in}(t) = 5 \cos(10^7 t) V$ for $-\infty < t < \infty$.
 b) Find $v_o(t)$ if $v_{in}(t) = 5 \cos(10^6 t) V$ for $-\infty < t < \infty$.



a) $\omega = 10^7 \text{ rps} \Rightarrow jX_L = j50 \Omega ; jX_C = -j50 \Omega$

$\therefore I = \frac{5}{50 \Omega + j50 \Omega - j50 \Omega} = \frac{5}{50 \Omega} = 0.1 \text{ A} \angle 0^\circ$

$V_o = I(-j50 \Omega) = -j5 \text{ V} = 5 \text{ V} \angle -90^\circ$

$\therefore v_o(t) = 5 \cos(\omega t - 90^\circ) = 5 \sin \omega t$

b) $\omega = 10^6 \text{ rps} \Rightarrow jX_L = j5 \Omega ; jX_C = -j500 \Omega$
 $Z = 50 + j(5 - 500) = 50 - j495 \Omega = 497.5 \angle -84.2^\circ$

$I = \frac{5 \angle 0^\circ}{497.5 \angle -84.2^\circ} = 10.05 \text{ mA} \angle 84.2^\circ$

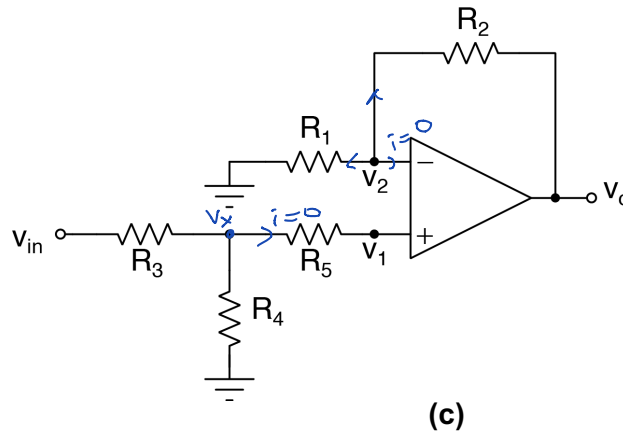
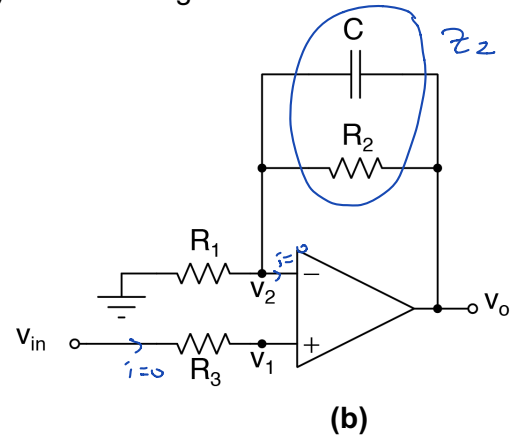
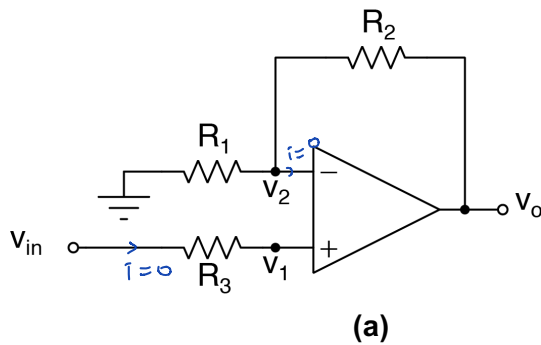
~~$V_o = I(jX_C) = I(-j500 \Omega) = I(500 \angle -90^\circ)$~~

$V_o = I(jX_C) = I(-j500 \Omega) = I(500 \angle -90^\circ)$
 $= 5.025 \text{ V} \angle -6^\circ$

$v_o(t) = 5.025 \cos(10^6 t - 6^\circ)$

Question 4 [25 points]:

Find the transfer function $H(\omega) = \frac{V_{out}}{V_{in}}$ for the following OPAMP circuits. Assume that the OPAMPs are ideal (i.e., $A \rightarrow \infty$, $R_{in} \rightarrow \infty$, $R_{out} = 0$). Show all your reasoning and derivations.



a) $V_1 = V_{in}$ (no current on R_3)
 $V_2 = \frac{R_1}{R_1 + R_2} \cdot V_o$ (voltage divider)

Using $V_2 = V_1 = V_{in}$ (assuming OPAMP is not saturated)

$$V_{in} = \frac{R_1}{R_1 + R_2} \cdot V_o \Rightarrow H(\omega) = \frac{V_o}{V_{in}} = \boxed{1 + \frac{R_2}{R_1}}$$

b) $V_1 = V_{in}$ (no current on R_3)
 $V_2 = \frac{R_1}{R_1 + Z_2} \cdot V_o$ (voltage divider)

Using $V_2 = V_1 = V_{in}$ (assuming OPAMP is not saturated)

$$V_{in} = \frac{R_1}{R_1 + Z_2} \cdot V_o \Rightarrow H(\omega) = \frac{V_o}{V_{in}} = 1 + \frac{Z_2}{R_1}$$

where $Z_2 = R_2 \parallel \frac{1}{j\omega C} = \left(\frac{1}{R_2} + j\omega C \right)^{-1} = \frac{R_2}{1 + j\omega R_2 C}$

$$\boxed{H(\omega) = 1 + \frac{R_2}{R_1} \cdot \frac{1}{1 + j\omega R_2 C}} \quad \text{OR, } H(\omega) = \left(\frac{R_1 + R_2}{R_1} \right) \left(\frac{1 + j\omega \frac{R_1 R_2}{R_1 + R_2} C}{1 + j\omega R_2 C} \right)$$

$$c) \quad V_1 = V_x \quad (\text{no current on } R_5)$$

$$= \frac{R_4}{R_3 + R_4} \cdot V_{in} \quad (\text{voltage divider})$$

$$V_2 = \frac{R_1}{R_1 + R_2} V_o \quad (\text{voltage divider})$$

Using $V_1 = V_2$ (assuming OPAMP is not saturated)

$$\frac{R_4}{R_3 + R_4} V_{in} = \frac{R_1}{R_1 + R_2} \cdot V_o$$

$$H(\omega) = \frac{V_o}{V_{in}} = \boxed{\frac{R_1 + R_2}{R_1} \cdot \frac{R_4}{R_3 + R_4}}$$